

Chapter: Metabolism (til 7.6)

1.1 metabolism and its types.

Metabolism:-

The word 'metabolism' means to change. The sum of all the chemical reactions going on within ~~the~~ each cell of the living organisms for maintenance and sustainability of life is known as Metabolism.

Metabolism is divided into two types i.e.:-

- * Anabolism
- * Catabolism

Catabolism:- Biochemical reactions which break down complex molecules to simpler molecules is known as catabolic reactions or Catabolism.

FOR EXAMPLE: ① Digestion is a catabolic process in which larger and complex food is broken down into simpler ~~glucose~~ molecule.

② Cellular respiration is also a catabolic process in which glucose molecule is broken down into CO_2 and H_2O by the series of all reactions in the body cells.

Anabolism :-

Chemical reactions in which build up from smaller molecule from smaller and simpler molecule is known as anabolic reactions or anabolism.

~~For Example :-~~
~~① Photosynthesis is an anabolic process in which glucose molecule is converted into CO_2 and H_2O~~

For Example :-
Photosynthesis is an anabolic process where simpler molecule like CO_2 and H_2O are converted into larger molecule glucose.

② Condensation of glucose molecule is to form glycogen in muscle and liver cell is also an anabolic reactions. Usually, Energy is released in catabolism, and is utilized in anabolism.

7.2 Enzymes and their characteristics :-

These metabolic reactions in living organisms sped up by specific proteins in the body is known as biological catalysts or enzymes.

Enzymes are biologically active globular proteins which are extremely speed up the biochemical reaction in the living organism.

Substrates:-

Enzymes act on a specific molecule in a cell is known as substrate. Substrates are the molecules entering in a chemical reactions. After binding to their specific enzymes, substrates undergo a chemical changes resulting in a new bonding arrangements b/w the molecules.

Products:-

The substrates modified by an enzyme is known as Products.

The enzymes, which stay inside the cell to speed up the reactions are known as intracellular enzymes.

E.g:- Enzymes of glycolysis working in the cytoplasm.

Other enzymes made inside the cell are allowed to go out of the cell to do their work is known as extracellular enzymes.

E.g:- Pepsin enzymes are working in the stomach cavity.

Fungi and bacteria release extracellular enzymes to digest their food. Study of enzymes, their structure, types and role of enzymes is known as enzymology.

2.1 Characteristics of enzymes:-

Globular Enzymes are globular proteins:-
Enzymes are made up of chains of amino acids to form a 3d proteins molecule known as globular proteins. Few enzymes are RNA called ribozymes.

Enzymes increase rate of reaction:-
Reactions may initiate without involvement of enzymes but it will be extremely slow and takes months or years to complete the reactions. Enzymes ~~sp~~ speed up the reactions millions of time faster ~~than as compared~~ as compared to non-enzymatic reactions. It is important to know that enzymes speed up the chemical reactions and do not start a reaction.

Enzymes are in small quantity:-
Structure of enzymes slightly change during the reaction but they regain their actual shape, so they can be unchanged at the end of the reaction. so they can be used repeatedly. A very small quantity of an enzyme is capable of catalyzing a huge amount of substrate.

Enzymes are specific:

Enzymes are usually very specific so catalyze only one type of reaction and it will not act on different substrate.
for example:

Amylase can only act on starch, and not on proteins or fats.

5. Enzymes require co-factor:

Many enzymes require as an additional non-protein components for their proper functioning are called co-factor.

Co-factor acts as a bridge b/w protein part of enzyme and its substrate. Co-factor may provide energy to enzyme. There are 3 types of co-factor.

① Activator:-

Enzymes require detachable inorganic ions called as co-factor, called activator such as Zn^{++} , Fe^{++} , Cu^{++} and chlorides etc.

FOR EXAMPLE:- Activity of salivary amylase is increased in the presence of chloride ions.

② Prosthetic group:-

If co-factor is an organic molecule tightly and permanently packed

attached to the enzyme it is known as prosthetic group.

FOR EXAMPLE :- flavin Adenine dinucleotides (FAD) & Haem group etc.

③ Co-enzyme :-

detachable if co-factor is an organic molecule it is called co-enzyme

EXAMPLES :-

NAD (nicotinamide Adenine dinucleotides), Co-enzyme A & vitamin A and C.

④ Regulation of enzyme production and activity :-

Synthesis of enzymes within the cell and can be increased or decreased by the cell according to requirements. Enzymes activity can be regulated by inhibitors or activators within or from out of the cell.

⑤ Enzymes make metabolic pathways.

Many enzymes work together & in a specific sequence to make metabolic pathways. A Metabolic pathway is series of connected chemical reactions where intermediate molecules are produced before conversion of an initial substrate into final product.

⑧ Energy of activation :-
minimum amount of extra energy to
required to activate the molecules to
~~activate~~ or react or start the reaction is
called energy of activation. Enzyme lowers
activation energy so that these reactions
can take place in the body temperature.
Enzymes lower activation energy by
changing the shape and charge of
substrate. Enzyme may provide different
mechanism of reaction to lower the
energy activation.

7.3 Mechanisms of Enzyme action.

Most enzymes are far larger than the
substrates they act. Specific part on the
globular surface of the enzyme where
substrates binds and actually take part in
the reaction. is known as active site. An
active site is usually a very small portion
of the enzyme and is the charge bearing
cavity having a specific shape.

* Specific substrates molecules temporarily
binds to the active site of an enzyme
to form enzyme substrate (ES) complex.

Once (ES) complex is formed, enzymes catalyze
the reaction to convert ES complex into
Enzyme Product (EP) complex. thus converting
ES complex into enzyme Product (EP) complex.

finally EP complex breaks up into products and enzymes. The enzymes molecules remains unchanged at the end of the reaction and is available again. If needed for the same reaction again.